

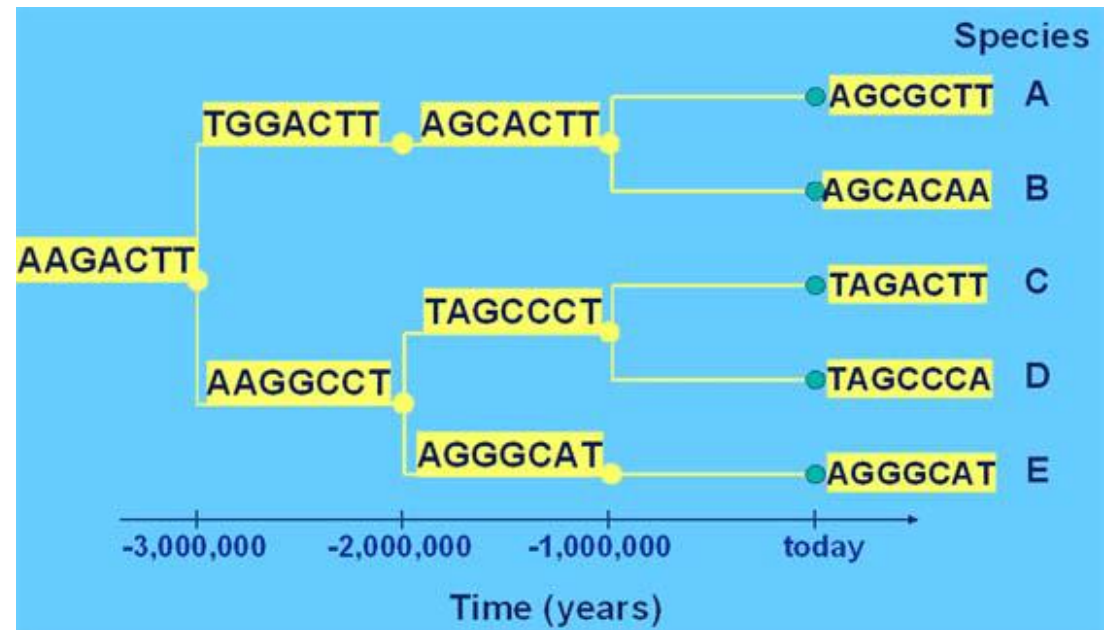
Phylogenetics - II

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The basic principle – Viewing the past, through modern eyes!

- All organisms contain DNA, RNA, and protein molecules
- Related organisms have a high degree of *similarity* in the sequences and structures of these molecules
- Distantly related organisms show a higher degree of *dissimilarity*
- Sequences change (mutate) over time as species diverge from each other
- We can study differences in the gene sequences to infer the evolutionary trajectories of many species

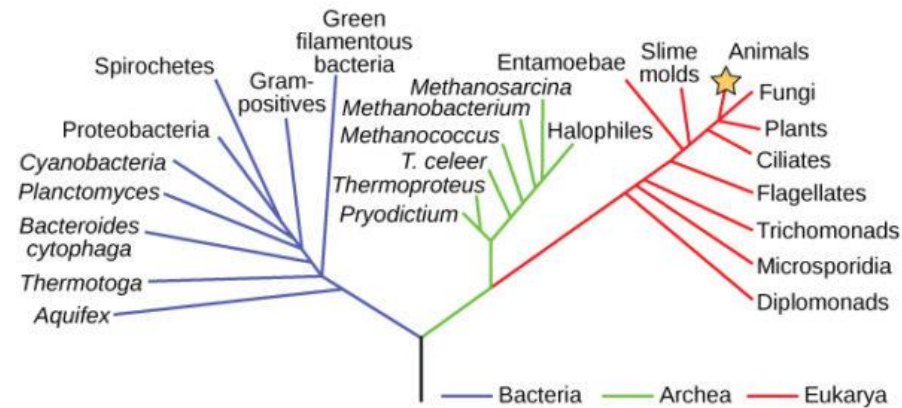


We are going back in time

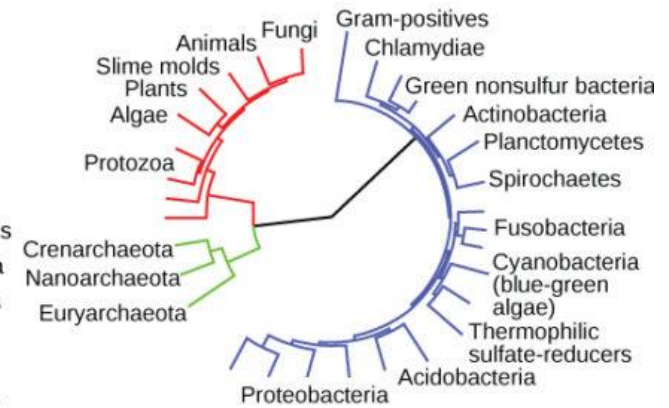


How can we represent evolutionary relationships between species?

Relationships are best illustrated by trees, which can be represented in multiple forms



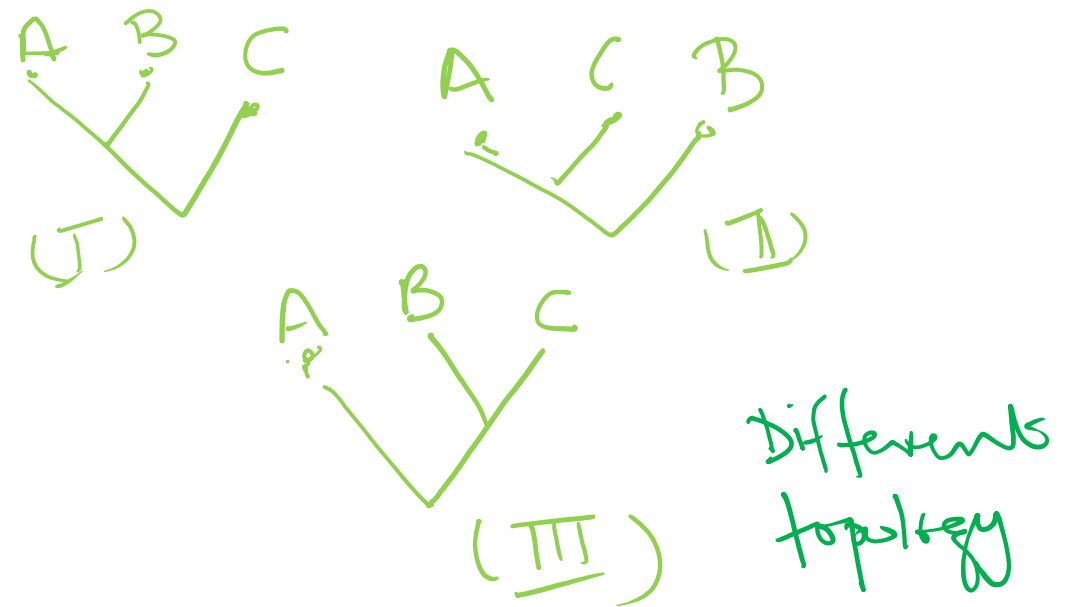
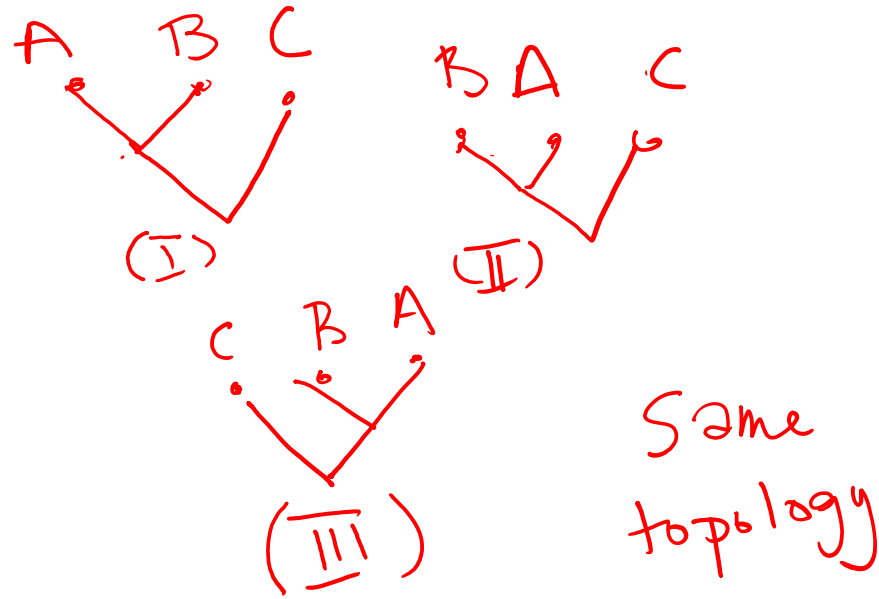
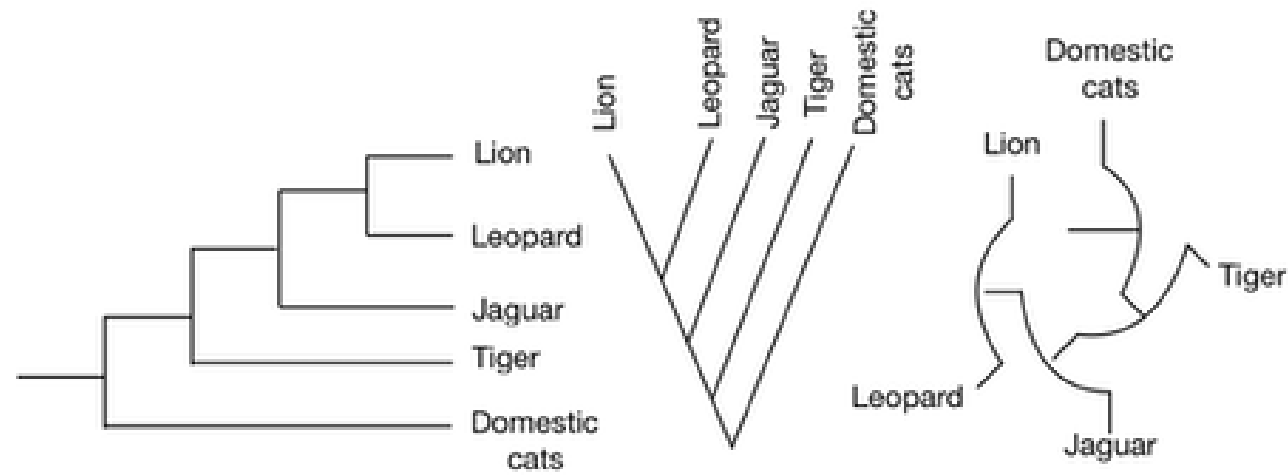
(a) Rooted phylogenetic tree



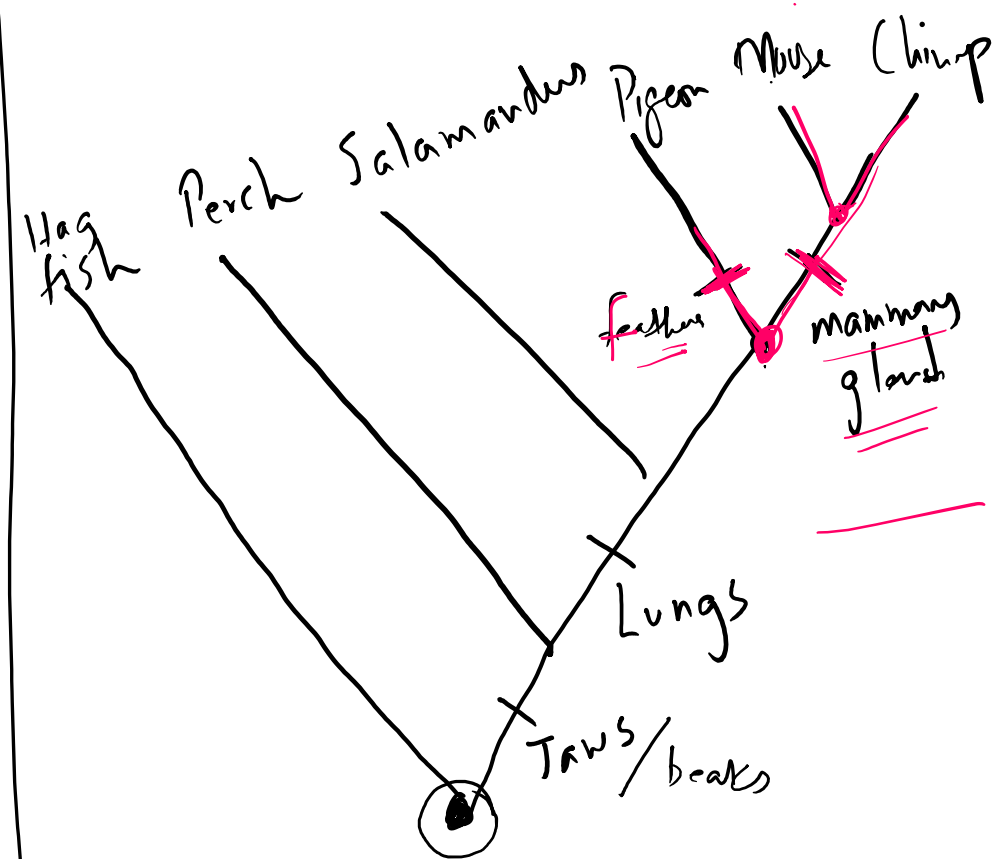
(b) Unrooted phylogenetic tree

The branching pattern is called **topology of the tree**

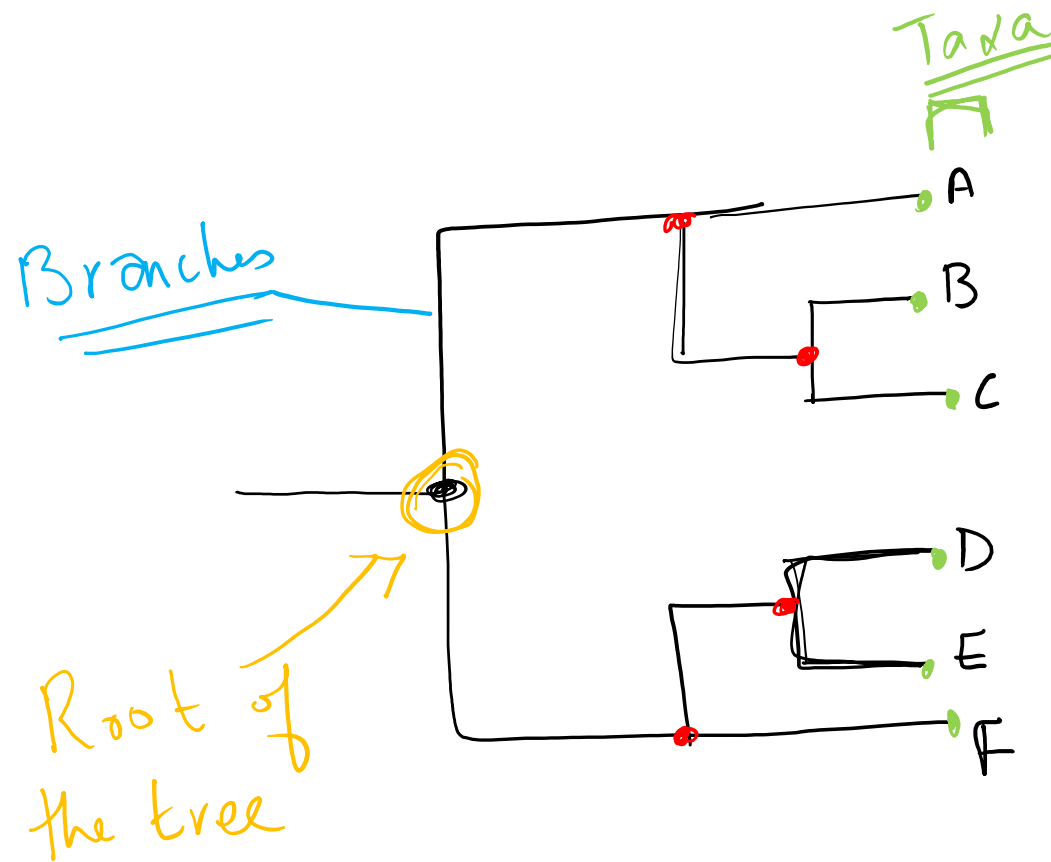




	Jaws	Lungs	Feathers	Mammary glands
Hag Fish	X	X	X	X
Perch	✓	X	X	X
Salamander	✓	✓	X	X
Pigeon	✓	✓	✓	X
Mouse	✓	✓	✓	✓
Chimp	✓	✓	✓	✓



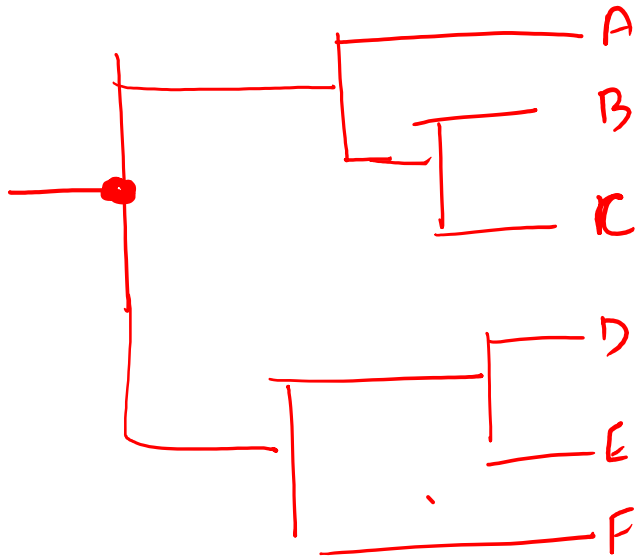
Tree terminologies



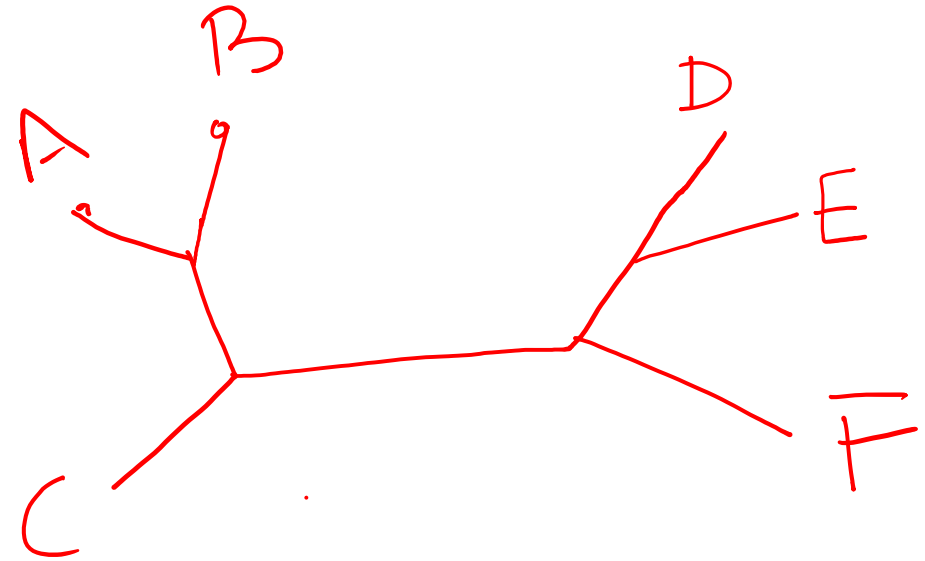
- Internal nodes
- Terminal nodes



Rooted vs Unrooted trees



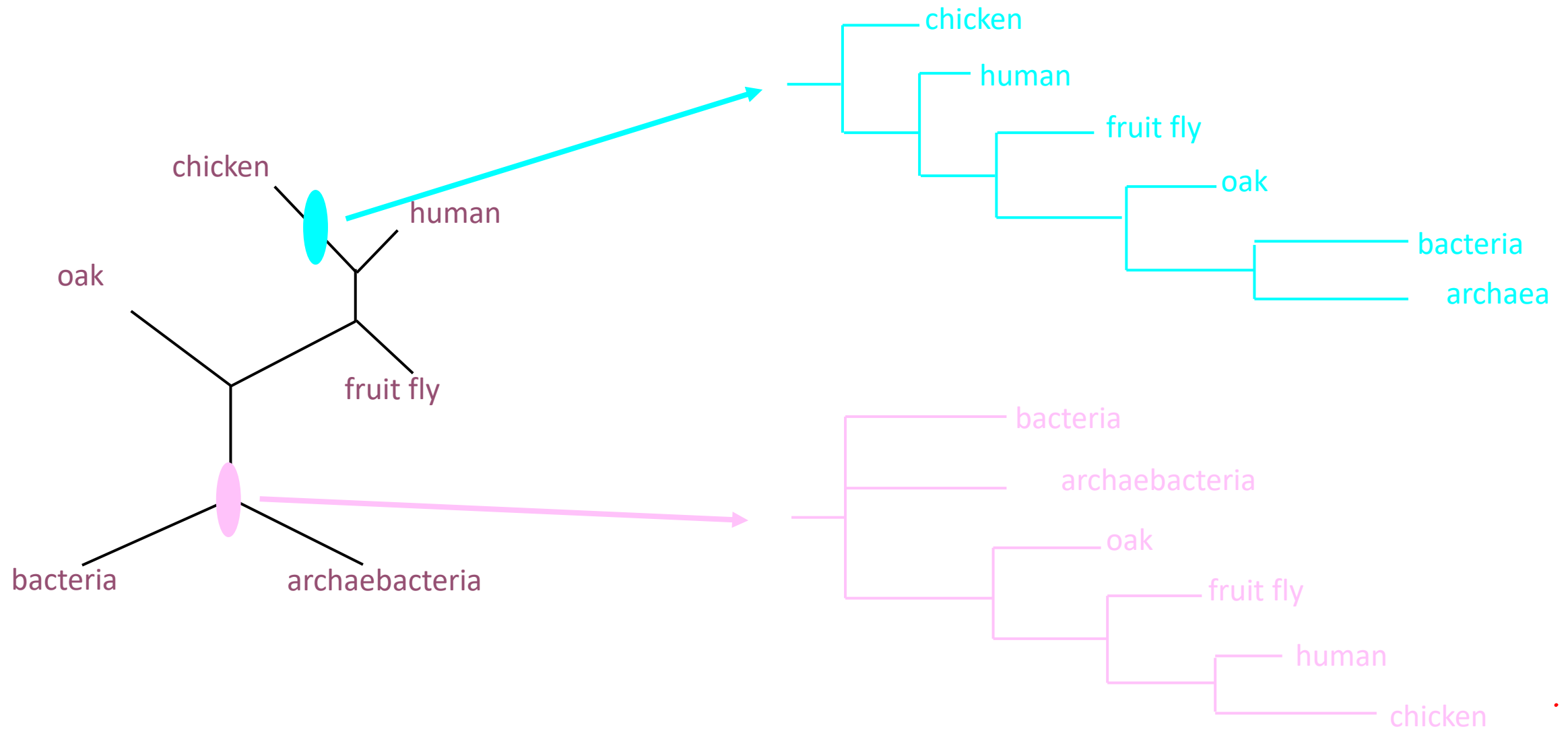
Rooted tree



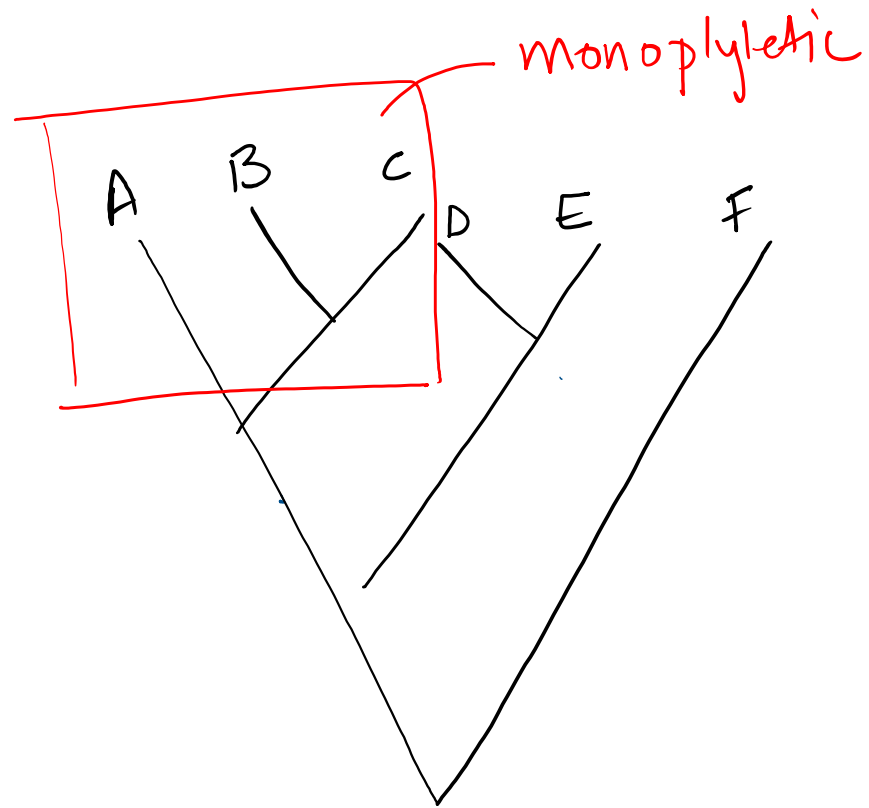
Unrooted tree



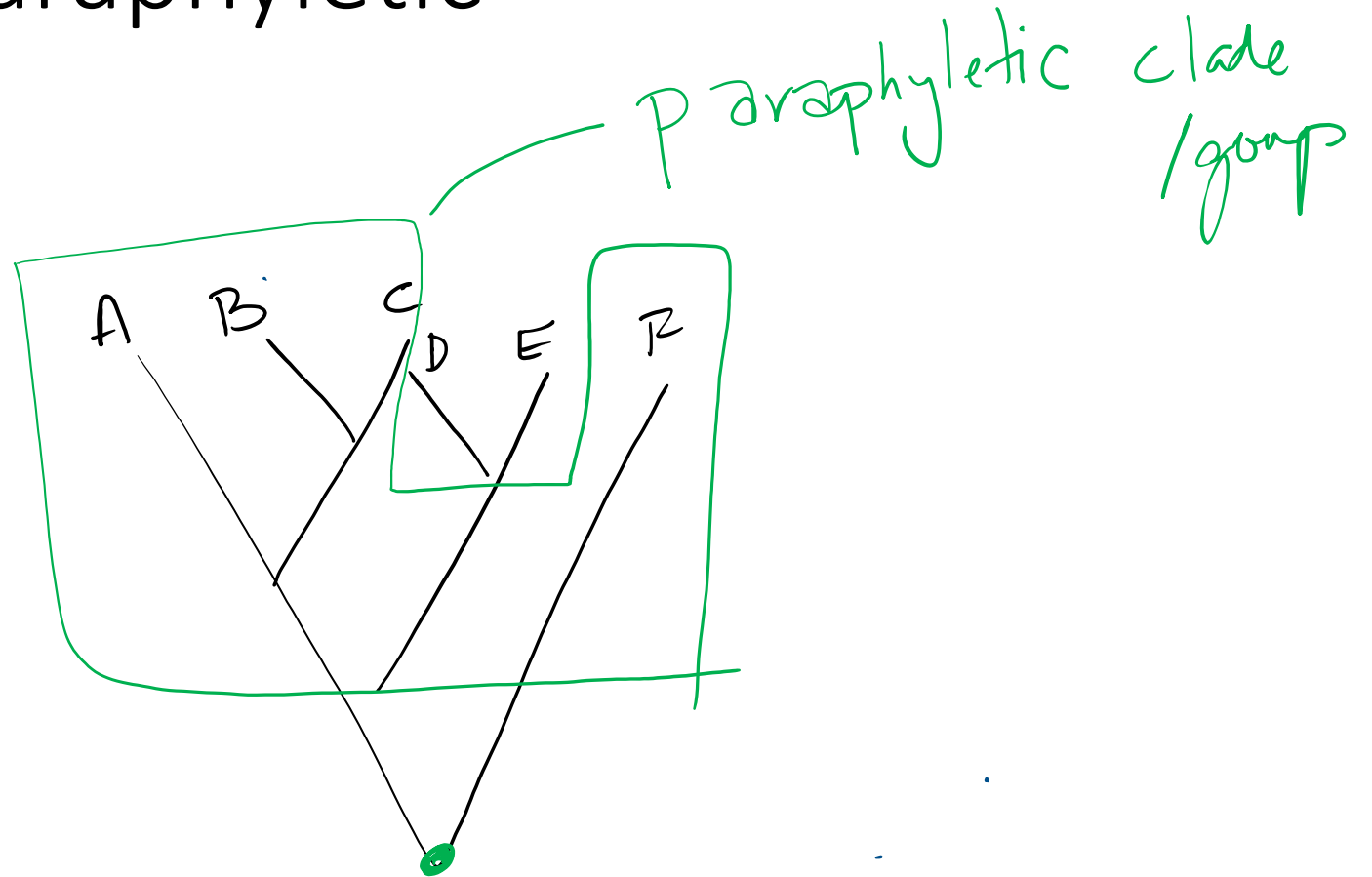
Rooting and tree interpretation



Monophyletic vs Paraphyletic



(I)



(II)



In short!

- ***What we need:*** gene/protein sequences (or any other form of molecular characters)
- ***What we want:*** compare these sequences across species
- ***How to present:*** we take help from the concept of trees
- Where,
 - ***Nodes*** represent taxa (our sampled species)
 - ***Branches*** represent the relationships between taxa
 - Branch lengths represent evolutionary time/changes since divergence

Phylogram
Cladogram

